Systemic Tau as a homeostatic trigger for structural self-repair in event-driven networks

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https://doi.org/10.5281/zenodo.23001549
27 September 2026
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Abstract

A network that replaces a missing connection requires a signal computed from events it already emits. Systemic Tau, τ_s, is the mean pairwise Kendall rank correlation τ_b of module activity on a common event index. Because the mean is signed, an increase in one pair and a decrease in another can cancel, and τ_s can remain inside a band calibrated on the intact network. The experiment therefore uses, as its operative alarm, the mean absolute change in τ_b on the edges present at the end of calibration, and it retains the signed mean as a control. The definition of τ_s is left unchanged. In a feed-forward network of seven modules that computes exclusive-or with six excitatory edges, three edges are deleted at trial 2,900, and accuracy is averaged over the following 1,500 trials. The comparisons, the window and the decision rules were fixed before seeds 1 to 30 were run. Before the lesion, mean accuracy on the matrix alarm is 0.917. Without structural repair, accuracy in the post-lesion window is 0.522. The same window on a network that is never lesioned remains at 0.848. The matrix alarm finishes at 0.786. It restores six edges in every seed, by exactly three repairs, and its first repair falls within 50 trials in 29 of 30 seeds. The mean rate of false alarms in the 400 trials that precede the lesion is 4.0 per 1,000 trials. The paired difference against repair directed at a random admissible target is 0.276 (95% interval 0.237 to 0.314). Against the same repairs placed at random times it is 0.216 (0.181 to 0.257). Against no repair it is 0.264 (0.223 to 0.299). Each lower bound exceeds +0.05. Against an alarm that uses the label error and the same restorative action, the difference is −0.019 (interval −0.036 to −0.001). The lower bound remains above the margin of −0.05, and the interval lies entirely below zero. When label polarity is reversed and the wiring is left intact, the event-driven conditions stay within a narrow range. An online long short-term memory network, damaged by a mask on its weights rather than by deletion of edges, relearns the task because training continues.

Introduction

When part of the wiring of a computing network is lost, the usual remedy is to retrain the weights that remain. Methods for continual learning are largely occupied with protecting that retraining from interference between successive tasks [12]. A different remedy is to let structure itself be the effector of a homeostatic loop [11]. Ashby’s question, for any particular system, is which variable is essential [1]. For a controller that must decide where a connection is to be put back, a useful essential variable is a concordance the modules can estimate from the events they already emit.

Systemic Tau is that concordance. It is the mean, over unordered pairs of modules, of Kendall's τ_b computed on a shared event index [2, 3]. The name, and the operational standard used by the reference implementation, are those set out in the earlier notes on this quantity [5, 6]. The coefficient is the cosine of the sign-difference vectors, with ties entered as zeros. The matrix of pairwise values is therefore a Gram matrix, and τ_s is bounded below by $-1/(N-1)$. For the seven modules used here the bound is $-1/6$. The same construction makes a limitation plain. Because the mean is signed, a lesion in a network that contains both cooperating and complementary streams can raise some pairs and lower others by similar amounts. The mean can then remain inside a band that was calibrated on the intact network.

The experiment asks whether an alarm built from this concordance can tell an event-driven exclusive-or network when, and where, to restore a deleted edge. The signed mean remains the definition of τ_s and remains one condition in the comparison. The operative alarm is the mean absolute change of τ_b on the edges present at the end of calibration. The window is 280 ticks. It was fixed before these seeds were run, as the shortest candidate among those specified in advance that is compatible with detection within 50 trials. A check on five separate tuning seeds, restricted to the period before the lesion and using circular-shift surrogates, adopted the same window for the signed mean [7]. That check records the behavior of the signed mean. It does not replace the window used by the edge alarm, and a signed mean that lies inside its surrogate null does not set the edge alarm aside.

Results

Lesion of three edges

The network has seven modules: four Bernoulli inputs that carry a complementary code of two bits, two hidden modules and one output. The initial wiring consists of the six excitatory edges that implement exclusive-or. Calibration occupies trials 2,000–2,499, inside one input regime. Trials 2,500–2,899 estimate false alarms in that same regime. At trial 2,900 the lesion deletes three of the six edges. A second input regime begins at trial 3,000. The primary endpoint is the mean prequential accuracy over the 1,500 trials that follow the lesion. The prediction on each trial is formed before any adaptation uses that trial.

Before the lesion the conditions agree. Mean accuracy is 0.917 on the matrix alarm and 0.919 on the condition that makes no structural repair. Four seeds, numbered 3, 12, 15 and 24, remain below 0.8 on the unrepaired condition in that pre-lesion window. The sensitivity analysis repeats the primary contrasts without them. After the lesion the conditions separate (Fig. 1 and Table 1). No repair falls to 0.522 and finishes with three edges. A yoked controller that replays the same three operation types and the same signs on random admissible targets also finishes with six edges, and its accuracy is 0.510. The edges are present, and they are the wrong edges. The matrix alarm finishes with six edges in every seed, after exactly three repairs, and its accuracy is 0.786. The network that is never lesioned, scored on the same later window and under the same change of input regime, reaches 0.848.

Three superiority contrasts meet the interval criterion fixed in advance (Fig. 1 and Table 2). The matrix alarm minus a random target is 0.276, with 95% interval 0.237 to 0.314 and paired standardized effect 2.50. Minus random time, the difference is 0.216, with interval 0.181 to 0.257 and effect 2.00. Minus no repair, the difference is 0.264, with interval 0.223 to 0.299 and effect 2.44. Each lower bound lies above +0.05. Holm-adjusted Wilcoxon p-values for these three contrasts fall below 10^{-8}. The decisions follow the interval rule. The rank test is reported with them and is not itself the rule.

The comparison with the error alarm is a test of non-inferiority. The difference is −0.019, with interval −0.036 to −0.001 and paired effect −0.38. The lower bound lies above −0.05, so the non-inferiority

criterion is met. The upper bound lies below zero, so the error alarm finishes higher, at a mean accuracy of 0.805. In one seed that alarm restores a single edge and then makes no further change to the graph, leaving four edges. The matrix alarm restores three edges in every seed.

The mean over 1,500 trials mixes the trials spent waiting for the repairs with the trials that follow them. The summaries that follow describe the time course inside that same window. They were not additional decision endpoints, and intervals were not computed for them. The first 50 trials after the lesion average 0.588 under the matrix alarm. The median seed receives its third repair 118 trials after the lesion. From the trial after that third repair to the end of the window, mean accuracy is 0.807. Over trials 200 through 1,499 after the lesion, the repaired network averages 0.800 and the unlesioned network averages 0.845. Once the three edges have been restored, most of the distance from the unrepaired value of 0.522 toward the unlesioned network has been closed. A residual gap remains.

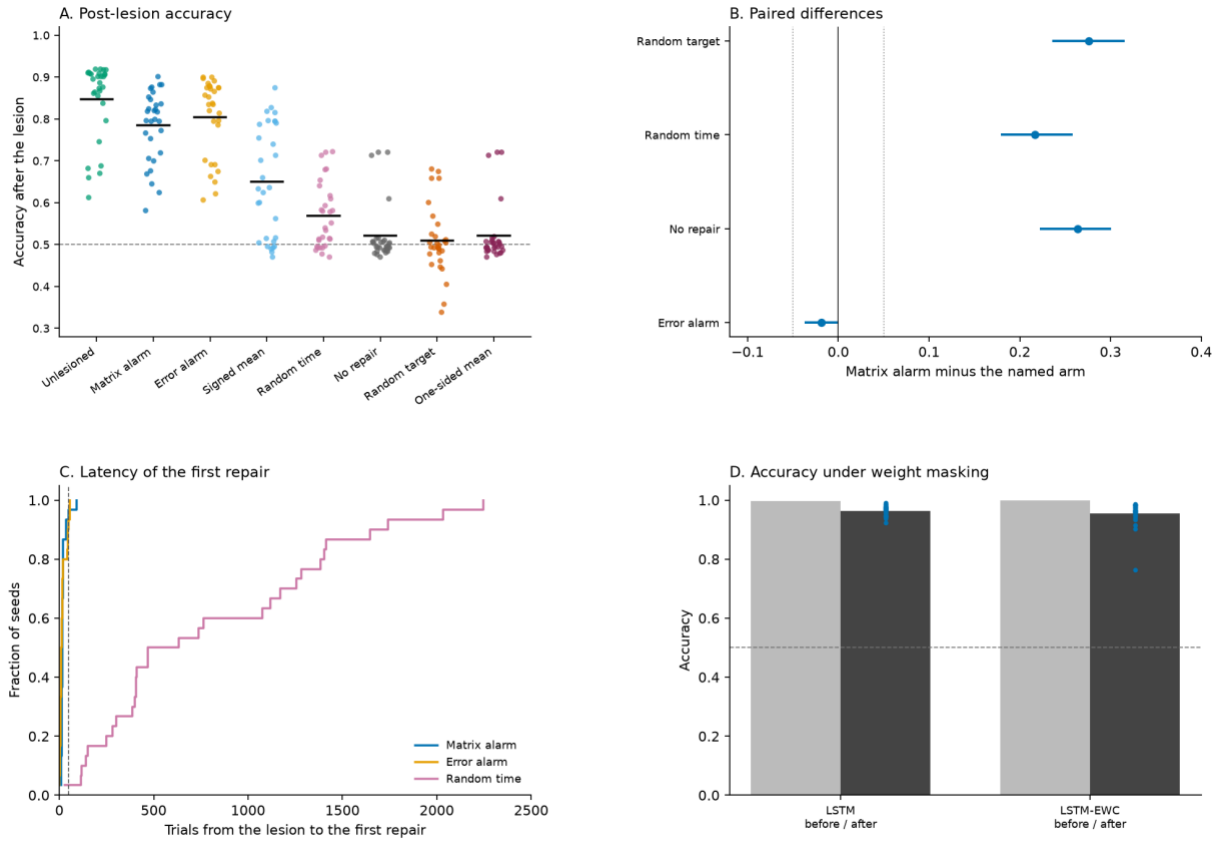


Figure 1. Accuracy, paired differences and repair latency for seeds 1 to 30. **(A)** Each point is the post-lesion accuracy of one seed. The black dash is the mean. The dashed horizontal line is chance, equal to 0.5. **(B)** Paired difference of the matrix alarm minus the named condition, with the 95% bias-corrected and accelerated interval. The solid vertical line is zero. The dotted lines are the margins +0.05 and −0.05. **(C)** Fraction of seeds whose first structural repair has occurred, plotted against trials since the lesion. The dashed vertical line marks 50 trials. **(D)** The long short-term memory reference receives a different intervention. Half of each weight matrix is set to zero and held at zero, and training continues. In each pair the left bar is the mean accuracy before that mask and the right bar is the mean after it. The blue points are the thirty seeds after the mask.

Table 1. Lesion experiment. Means over 30 seeds. Accuracy is prequential. The column "Within 50 trials" counts seeds whose first structural repair falls within 50 trials after the lesion. False alarms are counted per 1,000 trials on trials 2,500–2,899 and include alarms that do not change the graph. Edges are those present at the end of the run.

Condition	Before	After	Within 50 trials	False alarms per 1,000	Final edges
Unlesioned	0.919	0.848	0/30	0.0	6.0
Unlesioned, signed mean	0.918	0.833	0/30	1.9	6.0
Matrix alarm	0.917	0.786	29/30	4.0	6.0
Error alarm	0.919	0.805	28/30	1.8	5.9
Signed mean	0.918	0.650	4/30	1.9	5.4
Random time	0.916	0.569	1/30	0.9	5.6
No repair	0.919	0.522	0/30	0.0	3.0
Random target	0.919	0.510	29/30	0.0	6.0
One-sided signed mean	0.919	0.522	0/30	0.5	3.0

Table 2. Primary paired contrasts on post-lesion accuracy: matrix alarm minus the named condition. Intervals are bias-corrected and accelerated, from 10,000 resamples. Superiority requires the lower bound to exceed +0.05. Non-inferiority, used for the error alarm, requires the lower bound to exceed −0.05.

Contrast	Difference	95% interval	Decision
Random target	0.276	0.237 to 0.314	Superiority
Random time	0.216	0.181 to 0.257	Superiority
No repair	0.264	0.223 to 0.299	Superiority
Error alarm	−0.019	−0.036 to −0.001	Non-inferiority

Detection latency and the signed mean

The first repair under the matrix alarm falls within 50 trials in 29 of 30 seeds. The remaining seed, seed 11, has a pre-lesion accuracy of 0.940 and receives its first repair 91 trials after the lesion. It still ends with six edges and with a post-lesion accuracy of 0.800. The mean false-alarm rate is 4.0 per 1,000 trials, against a limit of 20. One seed reaches 20.0, and seven seeds have none. Both parts of the detection criterion are therefore met. The median latency of the first repair is 17 trials. Under random timing the median latency is 552 trials, and one seed in thirty is repaired within 50 trials. That condition finishes at 0.569, with 5.6 edges on average, because some of its allotted attempts fall before the lesion, find the wiring intact and do not change the graph.

After the three repairs, the deviation of the edge concordances often remains outside the calibration band. Those later crossings find the baseline wiring already complete. They add no edge, and they are not copied into the yoked conditions. The matrix alarm produces 38.4 such crossings on average over the run. The false-alarm rate does not include them, because they fall after the lesion. The refractory period of 20 trials still applies.

The signed mean detects this lesion poorly. That fact is why the signed mean was retained as a control, and why the definition of τ_s was not altered in order to avoid the cancellation. Post-lesion accuracy under the signed mean is 0.650. A repair falls within 50 trials in 4 of 30 seeds. In 27 of 30 seeds the signed mean already lies inside its calibration band at the trial of the first matrix repair. The one-sided condition, which responds only when τ_s falls below the lower edge of that band, never repairs. Its accuracy is 0.522,

and it finishes with three edges. The matrix alarm minus the signed mean is 0.135, with interval 0.086 to 0.193. Minus the one-sided condition the difference is 0.264, with interval 0.223 to 0.300. The baseline signed mean lies outside its surrogate null in 0.90 of the thirty seeds. That fraction is recorded with the window check. It is not a condition for accepting or rejecting the edge alarm.

The sensitivity analysis keeps the 26 seeds whose unrepaired pre-lesion accuracy is at least 0.8, and drops seeds 3, 12, 15 and 24. The four decisions are unchanged. Against a random target the difference is 0.293, with interval 0.253 to 0.332. Against random time it is 0.234, with interval 0.195 to 0.273. Against no repair it is 0.281, with interval 0.237 to 0.317. Against the error alarm it is −0.022, with interval −0.041 to −0.002, and the non-inferiority criterion is again met. The contrast that asks whether the restored edges are the deleted ones, and the contrast that asks whether the restoration occurs at the time of the lesion, both meet the superiority criterion.

Polarity reversal with the wiring left intact

The second experiment deletes no edges. Every 1,000 trials the label polarity of the second bit reverses, a change in the concept to be learned [10], while the input rates remain at their initial values. Adaptation, defined as accuracy on the first 200 trials of each new block, lies between 0.391 and 0.419 for the event-driven conditions (Table 3). Retention on the blocks that return to the original polarity lies between 0.747 and 0.792. The matrix alarm has no missing baseline edge to restore, so the restorative action used after the lesion does not fire. Permission, in this experiment, to add an admissible edge or a bridge does not open a separation of the size seen after the lesion. The paired difference in adaptation, matrix alarm minus no repair, is 0.023, with interval 0.011 to 0.037. The interval lies above zero and below the margin of +0.05 used for the lesion endpoint. The lesion result does not reappear under polarity reversal.

Table 3. Polarity reversal, with no edges deleted. Means over 30 seeds. Adaptation is accuracy on the first 200 trials after each block boundary. Retention is the same quantity on the blocks that return to the original polarity.

Condition	Adaptation	Retention
Matrix alarm	0.413	0.785
Error alarm	0.416	0.755
Signed mean	0.400	0.785
Random time	0.418	0.779
No repair	0.391	0.792
Random target	0.419	0.747

Weight masking in a recurrent reference

The long short-term memory reference has 16 hidden units and takes one Adam step after each trial, at a learning rate of 0.01 [9]. At the lesion trial, half of every weight matrix is set to zero and held at zero. Biases are left intact. This intervention is not the deletion of three edges, and the accuracies in this section are not a ranking against Table 1. Before the mask, accuracy is 0.997. After the mask, with training continued, the post-lesion mean is 0.964. Elastic weight consolidation, with penalty strength $\lambda = 10$, finishes at 0.955 (Fig. 1D) [8]. The paired difference of the matrix alarm minus the recurrent network, scored on the post-lesion window of the edge experiment, is −0.178, with interval −0.211 to −0.153. Minus the consolidated network the difference is −0.170, with interval −0.203 to −0.143. The intervals lie below zero because the recurrent network is scored after a differently shaped intervention and because it continues to update its weights.

The value $\lambda = 10$ was selected before seeds 1 to 30 were used for the recurrent conditions. On tuning seeds 1,001 to 1,005, mean retention under polarity reversal was 0.765 at $\lambda = 10$, 0.726 at $\lambda = 100$ and 0.664 at $\lambda = 1{,}000$. The largest of these means is the value 10. On the reported polarity reversal, the consolidated network reaches an adaptation of 0.742 and a retention of 0.762. The network without consolidation reaches 0.660 and 0.687. Continued updates of the weights follow a reversed label. Restoration of a wiring that was never removed does not install the reversed function. The two experiments concern different effectors.

Table 4. Recurrent reference, means over 30 seeds. In the lesion column, half of each weight matrix is zeroed. The polarity-reversal columns use the same label reversal as Table 3, with no deletion of edges. The penalty strength of the consolidated network is $\lambda = 10$.

Condition	Lesion, before	Lesion, after	Polarity, adaptation	Polarity, retention
Long short-term memory	0.997	0.964	0.660	0.687
With consolidation, λ = 10	0.999	0.955	0.742	0.762

Discussion

On this seven-module, event-driven exclusive-or, an alarm that follows the absolute concordance of the baseline edges, and that restores the missing edge with the largest drop, puts the deleted wiring back. A random target that receives the same number of operations, of the same types and the same signs, does not restore the function. It finishes with six edges and with accuracy 0.510. The same attempts, spread across the run at random times, restore the function late and incompletely. No repair leaves accuracy at 0.522. The contrast on the identity of the restored edge and the contrast on the time of the restoration both meet the superiority criterion. The alarm that uses the label finishes higher by 0.019, and that difference remains inside the margin of 0.05.

The signed mean is a weak detector of this particular lesion. Cancellation is a property of the mean of signed concordances. The experiment does not redefine τ_s in order to remove that property. The matrix alarm is a second functional of the same pairwise coefficients, restricted to the edges the network possessed at calibration. Original modules are never removed. In the lesion experiment the controller does not add an edge that the calibration record did not contain.

Two limits are visible in the same runs. After the third repair, accuracy through the end of the post-lesion window is 0.807, short of the late-window accuracy of 0.845 on the network that was never lesioned. When the task changes by a reversal of label polarity and the wiring is left intact, the event-driven conditions remain close to one another. Homeostatic restoration of a known wiring answers the loss of that wiring. It does not, by itself, discover a new input–output function. The recurrent reference, which is permitted to keep changing its weights, follows the reversed label, and it does so under an intervention that is not a deletion of edges. The positive result on the lesion has the scope of the test that produced it: one feed-forward graph, one task, six edges, and a controller allowed to replace edges it has stored.

Methods

Task, modules and timeline

Each trial draws bits b_1 and b_2 independently and with equal probability, and takes as its label the exclusive-or of the two bits. The trial lasts 14 ticks. Four input modules carry the code (b_1, $1 - b_1$, b_2, $1 - b_2$) and emit independent Bernoulli events, with an on-rate of 0.70 and an off-rate of 0.02 in the first

regime, and an on-rate of 0.55 and an off-rate of 0.05 in the second. The output predicts 1 when its spike count on ticks 3 through 13 is at least 2. Hidden and output modules are discrete-time leaky integrate-and-fire units, with leak 0.3, synaptic weight of magnitude 1 and Gaussian noise of standard deviation 0.05. A spike resets the membrane. Edges are feed-forward. The initial edges, all excitatory, are I1→H1, I4→H1, I2→H2, I3→H2, H1→O and H2→O. Initial thresholds are drawn uniformly from the interval [1.2, 2.2] for the hidden modules and from [0.6, 1.4] for the output, and are bounded to [0.3, 5.0]. A separate search over thresholds proposes one perturbation of standard deviation 0.1 every 50 trials and keeps the perturbation when the Kendall score does not fall. A rejected proposal decays toward the score of the latest block with weight 0.2.

A run contains 6,000 trials, in blocks of 1,000. Calibration is trials 2,000–2,499. In the lesion experiment the deletion occurs at trial 2,900 and removes the floor of half the existing edges, which is three edges. The post-lesion window is the next 1,500 trials. The second input regime begins at trial 3,000. In the polarity-reversal experiment the rates remain at the values of the first regime, the label polarity of b_2 alternates by block, and no edge is deleted.

Concordance, alarm and action

Activity is read at a bin width of one tick. On a window of $W = 280$ ticks ending at the current trial, τ_b between two modules is the cosine of their sign-difference vectors. Systemic Tau is the mean of τ_b over the unordered pairs among modules that were alive at calibration. The operative deviation is the mean, over the edges present at the end of calibration, of the absolute change in τ_b from the calibration mean of that edge. The calibration record is stored before the controller acts, and it includes the sign of each edge. The alarm fires when the deviation exceeds its calibration mean by three standard deviations, with the standard deviation floored at 10^{-6}. The refractory period is 20 trials.

In the lesion experiment the action restores, with the stored sign, the missing baseline edge whose absolute concordance has fallen the most. If every baseline edge is already present, the action is empty. An empty action is counted as a false alarm when it falls in the pre-lesion window. It sets the refractory period. It is not entered in the list of repairs, and the yoked conditions do not receive it. Detection latency is the trial of the first repair that changes the graph at or after the lesion. In the polarity-reversal experiment, where the baseline edges are still present, the controller may add an admissible edge or a bridge. Original modules are never removed.

The signed-mean condition fires when τ_s leaves a two-sided calibration band of half-width three standard deviations. The one-sided condition fires only when τ_s falls below the lower edge of that band. The error alarm fires when the running error exceeds its own calibration band, and it uses the labels. Its structural action is the same restoration used by the matrix alarm. The unlesioned conditions skip the deletion.

Controls, seeds and decision rules

The matrix alarm is the primary condition and is run first. The random-target condition replays each recorded operation type and sign on a random admissible pair. The random-time condition places the same number of attempts independently and uniformly between the end of calibration and the end of the run, without the refractory spacing. Only operations that change the graph enter that number. Streams for the task, the lesion, the noise, the initial thresholds, the threshold search, the controller, the surrogates and the recurrent network are drawn from a seed sequence keyed by the seed and by the date 27 September 2026. The conditions of one seed therefore share inputs, noise and deleted edges.

Seeds 1 to 30 were reserved for the reported comparisons and were used once. Tuning seeds 1,001 to 1,005 were used for the window check and for the grid over the consolidation penalty, and they do not enter the reported contrasts. Arms are paired by seed. Each primary contrast is a mean paired difference, with a 95% bias-corrected and accelerated bootstrap interval from 10,000 resamples and a 90% interval used for the equivalence check [4]. A Wilcoxon signed-rank p-value is adjusted by the Holm procedure across the four primary contrasts. Superiority is met when the lower 95% bound exceeds +0.05, and is rejected when the 90% interval lies inside −0.05 to +0.05 or the upper 95% bound is negative. Otherwise the contrast is inconclusive. Non-inferiority is met when the lower 95% bound exceeds −0.05, and is rejected otherwise. The detection criterion is met when the first repair falls within 50 trials in more than half of the seeds and the mean false-alarm rate is at most 20 per 1,000 trials. Fewer than half of the seeds rejects the criterion. Exactly half, or a mean above 20, leaves it inconclusive. The design was sized so that a paired t test at $n = 30$ has a detectable standardized effect of about 0.53 at $\alpha = 0.05$ and about 0.64 at $\alpha = 0.0125$, for a test against zero. The interval rule is stricter. A paired standard deviation of 0.10 requires a mean difference of about 0.087 before the lower 95% bound clears +0.05.

The sensitivity analysis repeats the four primary contrasts on the seeds of the lesion experiment whose unrepaired pre-lesion accuracy is at least 0.8. The window check evaluates candidate windows in a fixed order on the tuning seeds and on pre-lesion data, and it adopts the first candidate whose baseline signed mean lies outside a surrogate null in at least 80% of those seeds. The adopted candidate was a window of 280 ticks and a bin width of 1, with pass fraction 1.0. The candidate with a window of 140 ticks and a bin width of 1 had pass fraction 0.4. The check does not replace the window of 280 ticks.

Recurrent reference

The long short-term memory network and its consolidated variant are secondary conditions [8, 9]. The consolidation penalty is $(\lambda/2)$ times a squared distance from an anchor, weighted by a diagonal empirical Fisher information. The Fisher information is estimated on the last 200 trials of each block of 1,000 trials and is then accumulated. The penalty strength is the grid value 10, selected as the maximum of the mean retention under polarity reversal on the five tuning seeds. Multiply–accumulate counts are stored for each recurrent seed. Synaptic events, module updates and pairwise evaluations of τ_b are stored for each event-driven seed. The two counts are different units and are not placed on one scale.

Data and code availability

This article is archived at https://doi.org/10.5281/zenodo.23001549. The reference implementation, version 0.1.2.dev0, together with the summary of each seed, the analysis, the window check, the selection of the consolidation penalty and the trial-level records, is archived at https://doi.org/10.5281/zenodo.23001551 and is developed at https://github.com/johelpadilla/tri-structural-repair. The configuration file used for the reported run has SHA-256 9451481deb61b28d26c9dbde0ea899b9163324a65c3538fe1fd1514a4ce16550. The event-driven runs began on 27 September 2026 at 14:29:32 (UTC−4). The penalty strength used for the consolidated recurrent condition is 10, as recorded in the selection file for that grid. The configuration file retains the value 100, which is the value written before the selection.

Acknowledgements

Language-model tools assisted the preparation of the text and the execution of the reference implementation. The author compared the reported values with the saved analysis and is responsible for the content.

Author contributions

J.P.-V. conceived the study, specified the experiment, carried out the analysis and wrote the manuscript.

Competing interests

The author declares no competing interests.

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